

PFOTM (Powerful Fisioterapis of the Month) Ranking System

Executive Overview

This specification document outlines the technical requirements, domain logic, and architectural guidelines for migrating the legacy manual **Powerful Fisioterapis of the Month (PFOTM)** ranking spreadsheet into a fully automated web application. The legacy workbook serves as an aggregation engine tracking individual performance metrics—such as session volume, attendance, discipline, package sales, and internal marketing referrals—to derive weighted scores and monthly rankings.

The primary objective of this project is to eliminate spreadsheet calculation vulnerabilities, establish dynamic rule-based scoring, automate historical analytics reporting, and deliver a gamified user interface for real-time performance visibility.

1. Legacy Spreadsheet System Architecture

The existing legacy system operates across four primary workbook components, functioning as raw data trackers, a point-calculation engine, an output leaderboard, and a historical KPI archive.

1.1 Aggregation Trackers (**Kunjungan**, **Rujukan SM**, **Keuangan**)

- **Operational Role:** These sheets log operational activity throughout the month.
- **Kunjungan (Visits):** Logs individual patient service sessions across specific clinical packages and appointment categories (e.g., **PAKET 1**, **PAKET 2**, **SESI NEW**, **TERAPI AWAL**). Generates a cumulative **Grand Total** visit count per clinician.
- **Rujukan SM (Social Media / Internal Referrals):** Logs patient acquisition originating from therapist-driven internal referrals, mapping referring staff (**Pengirim**) to total referral counts.
- **Keuangan (Financial Sales):** Logs direct revenue creation by attributing high-value treatment package sales to the responsible physiotherapist.

1.2 Point Scoring Engine (Data)

The core calculation layer ingests aggregated raw metrics and applies programmatic multipliers or deduction penalties to compile total performance points (Total Poin).

Performance Category	Source Sheet Field	Metric Type	Default Multiplier / Weight	Mathematical Impact
Attendance	Hadir	Operational Days	+1.0 per day	Base positive score expansion
Punctuality Penalty	Terlambat	Event Frequency	-1.0 per incident	Direct point deduction
Absence Penalty	Izin / Alpha	Event Days	Non-weighted / Deduction	Base availability penalty
Total in Clinic Visits	Jumlah Kunjungan	Session Volume	+1.0 per session	Volume-based baseline
Package Sales (Clinic)	Paket 1	Sales Unit	+3.0 per package	High-value driver weighting
Package Sales (Clinic)	Paket 2	Sales Unit	+6.0 per package	
Package Sales (Clinic)	Paket 3	Sales Unit	+12.0 per package	
Total Home Care Visits	TA/Sesi Visit	Session Volume	+2.0 per session	
Package Sales (Home Care)	Paket Visit	Sales Unit	+10.0 per package	
Internal Referrals	Rujukan SM	Patient Units	+2.0 per referral	Acquisition bonus weighting

1.3 Leaderboard Output (Rank)

- **Operational Role:** Serves as the executive summary and visual ranking board.
- **Structure:** Evaluates Nama, cumulative Total Poin, and computes ordinal position (Ranking).
- **Sorting Logic:** Implicit continuous evaluation where $\text{Rank} = f(\text{Total Poin})$ in strictly descending order.

1.4 Historical Analytics & KPI Engine (Rekap Data)

- **Operational Role:** Stores monthly historical logs to calculate longitudinal human resource key performance indicators (KPIs).
- **Calculated KPI Formulas:**

- **Attendance Rate (%):**

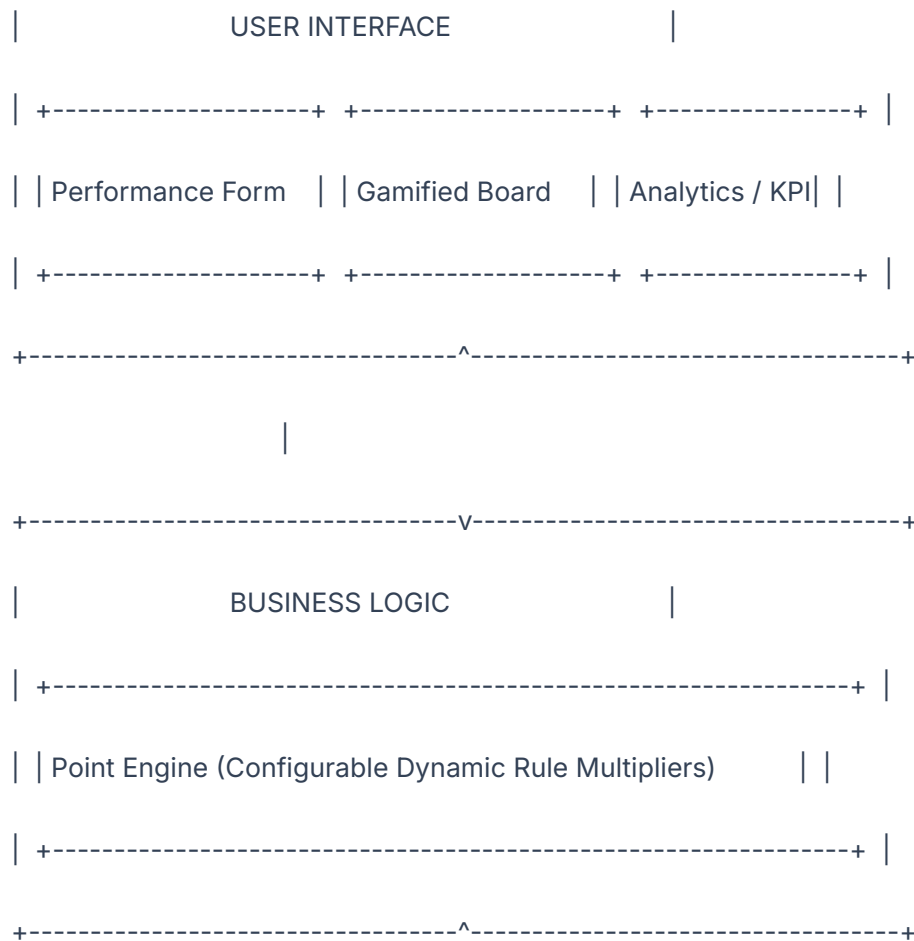
$$\% \text{ Kehadiran} = \left(\frac{\text{Hadir}}{\text{Hari Kerja Total}} \right) \times 100\%$$
- **Discipline Rate (%):**

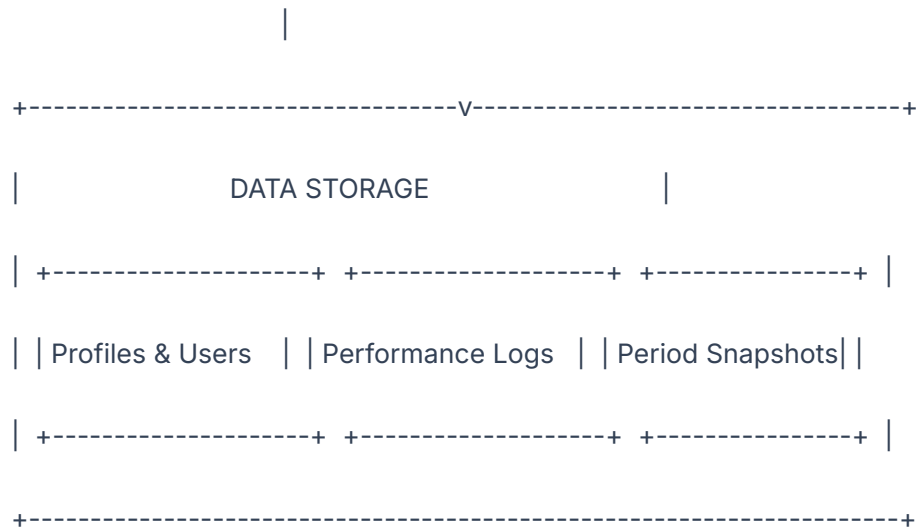
$$\% \text{ Disiplin} = \left(\frac{\text{Hadir} - \text{Terlambat}}{\text{Hadir}} \right) \times 100\%$$
- **Efficiency Ratio:**

$$\text{Kunjungan per Hadir} = \frac{\text{Total Visits}}{\text{Hadir}}$$

2. Technical System Requirements & Data Architecture

To ensure operational continuity and maintainability, the system must adhere to a modular architecture consisting of five core backend and frontend modules.





2.1 Profile & Resource Module

- Maintains a centralized directory of active, inactive, and historical physiotherapists.
- Attributes include: `Employee_ID`, `Full_Name`, `Role`, `Employment_Status`, `Base_Branch_Location`.

2.2 Performance Data Capture Module

- Replaces multi-sheet input with a single consolidated form supporting daily, weekly, or monthly inputs.
- **Input Schema Validations:**
 - All raw numbers must enforce non-negative integer constraints (≥ 0).
 - System must auto-calculate working days in period (`Hari Kerja`) while allowing HR manual override.

2.3 Dynamic Point Engine (Calculation Core)

- **Decoupled Weight Configuration:** Multipliers must **never be hardcoded**. Rules must be fetched dynamically from a backend `Point_Rules` configuration table.
- **Master Calculation Formula:**

$$\text{Total Points} = \sum_{i=1}^n (\text{RawMetric}_i \times \text{Weight}_i)$$

$$\text{Total Points} = (H \cdot W_H) + (T \cdot W_T) + (V \cdot W_V) + (P_1 \cdot W_{P1}) + (R \cdot W_R)$$

Where:

- $H = \text{Days Present}$, $W_H = \text{Attendance Weight}$
- $T = \text{Times Late}$, $W_T = \text{Late Penalty Weight (typically negative)}$
- $V = \text{Total Visit Count}$, $W_V = \text{Visit Weight}$
- $P_1 = \text{Package 1 Sales}$, $W_{P1} = \text{Package 1 Weight}$
- $R = \text{Referral Count}$, $W_R = \text{Referral Weight}$

2.4 Leaderboard & Visual Display Module

- Sorts all active profiles dynamically by **Total Points** descending.
- **Tie-Breaking Engine Logic:** In the event of equal **Total Points**, ordinal position is automatically calculated via secondary and tertiary parameters:
 - a. Higher total package sales (P_1).
 - b. Lower late incidents (T).
 - c. Higher discipline rate ($\% \text{ Disziplin}$).

2.5 Period Lock & Historical Reporting

- **Snapshot Lifecycle:** Provides a formal "End-of-Month Lock" action triggered by HR administrators.
- **Immutability:** Locking freezes the raw inputs, rule parameters, calculated scores, and ranks into a write-once historical record.
- **Automated Analytics Engine:** Automatically runs historical KPI calculations ($\% \text{ Kehadiran}$, $\% \text{ Disziplin}$, $\% \text{ Efficiency}$) on locked records.

3. Developer Implementation Guidelines

3.1 Database Schema Reference

-- User and Profile Directory

```
CREATE TABLE employees (
```

```
    employee_id VARCHAR(50) PRIMARY KEY,
```

```
    full_name VARCHAR(100) NOT NULL,
```

```
is_active BOOLEAN DEFAULT TRUE,  
  
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
  
);
```

-- Dynamic Rules Configuration Table

```
CREATE TABLE point_rules (  
  
    rule_id VARCHAR(50) PRIMARY KEY,  
  
    metric_key VARCHAR(50) UNIQUE NOT NULL,  
  
    metric_label VARCHAR(100) NOT NULL,  
  
    default_weight DECIMAL(5,2) NOT NULL,  
  
    is_penalty BOOLEAN DEFAULT FALSE,  
  
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
  
);
```

-- Period Tracking and Locking Engine

```
CREATE TABLE evaluation_periods (  
  
    period_id VARCHAR(7) PRIMARY KEY, -- e.g., '2026-03'  
  
    working_days INT NOT NULL,  
  
    is_locked BOOLEAN DEFAULT FALSE,  
  
    locked_at TIMESTAMP NULL  
  
);
```

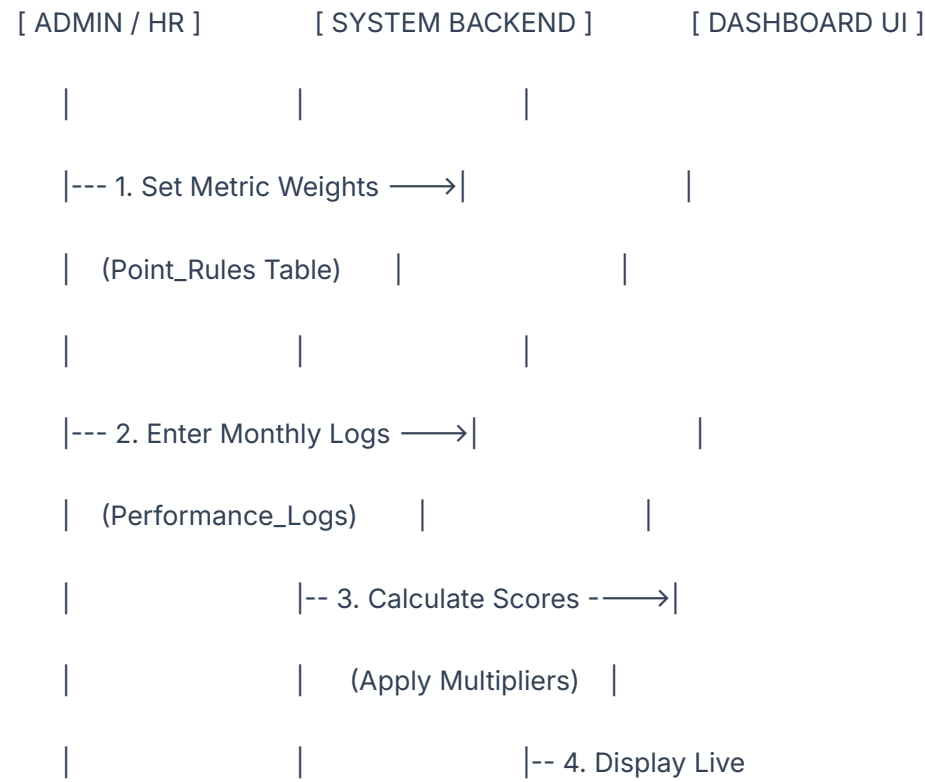
-- Raw Performance Entries

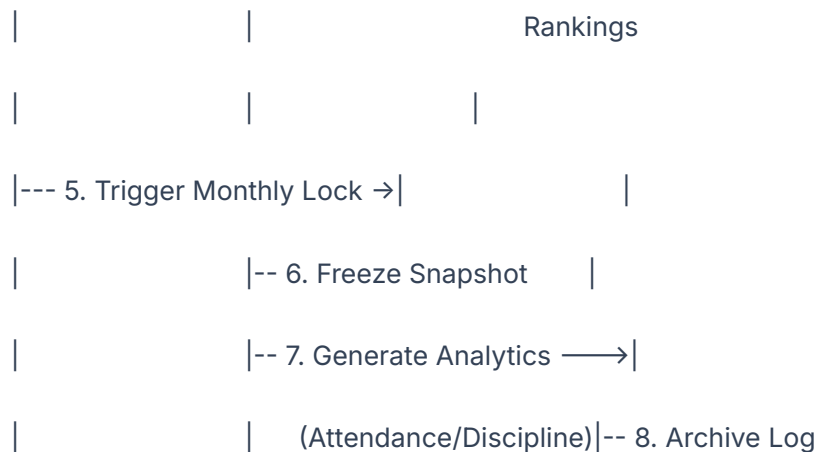
```
CREATE TABLE performance_logs (  
  

```

```
log_id SERIAL PRIMARY KEY,  
  
period_id VARCHAR(7) REFERENCES evaluation_periods(period_id),  
  
employee_id VARCHAR(50) REFERENCES employees(employee_id),  
  
days_present INT DEFAULT 0,  
  
times_late INT DEFAULT 0,  
  
total_visits INT DEFAULT 0,  
  
package_1_sales INT DEFAULT 0,  
  
referrals_sm INT DEFAULT 0,  
  
UNIQUE(period_id, employee_id)  
  
);
```

3.2 Key System Workflows





1. **Rule Initialization:** Administrative users configure current metric values via an Admin Panel.
2. **Data Ingestion:** Operational data is entered continuously or batched at period end.
3. **Real-Time Processing:** The system dynamically recalculates current standings for unlocked evaluation periods using the operational formula.
4. **Closing & Archiving:** HR reviews the monthly draft standings, resolves any discrepancies, and clicks "**Lock Period**". The system generates frozen analytics rows, permanently recording historical standings.

3.3 UI/UX Design Directives

- **Leaderboard Gamification:** The top 3 ranked physiotherapists must be visually distinguished using distinct UI cards, badge counters, or podium components.
- **Input Validation:** Form inputs must enforce native numeric type constraints, block negative numbers, and feature real-time live preview score calculations prior to form submission.
- **Audit Safety:** Any modification to a locked period requires explicit super-admin override privileges and logs an entry in an audit trail.